

SDP as a tool for quantum information

Lecture 2: Dual certificates, strong duality, and numerical SDPs

Marco Túlio Quintino

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SORBONNE
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Lecture 2: Dual certificates and computation

6. Duality theory 2.0

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7. Analytical applications

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- 10. Outlook

Recap

Primal points achieve; dual points certify

PRIMAL – achievable

$$\begin{aligned} \alpha &:= \sup_{\rho} \operatorname{tr}(A\rho) \\ \text{subject to } & \Lambda(\rho) = B \\ & \rho \succeq 0 \end{aligned}$$

DUAL – upper bound

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Feasible points are attainable
strategies

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$$\begin{aligned} \beta &:= \inf_Y \quad \text{tr}(BY) \\ \text{subject to} \quad &\Lambda^\dagger(Y) \succeq A \\ &Y = Y^\dagger \end{aligned}$$

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$$\alpha \leq \beta \quad (\text{weak duality})$$

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Linear objective + affine normalisation + PSD constraints $\implies$ SDP

6. Duality theory 2.0

From upper bounds to strong duality

The dual problem for state discrimination

PRIMAL – achievable

$$p_{\text{succ}}^* := \max_{\{M_i\}} \sum_{i=1}^N p_i \text{tr}(M_i \rho_i)$$

subject to $M_i \succeq 0 \quad \forall i$

$$\sum_{i=1}^N M_i = \mathbb{1}$$

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Write the Lagrangian and factorise M_i

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subject to $M_i \succeq 0, \quad \sum_i M_i = \mathbb{1}$

$$L(\{M_i\}, Y, \{\sigma_i\}_i) := \sum_i p_i \text{tr}(M_i \rho_i) + \text{tr} \left[Y \left(\mathbb{1} - \sum_i M_i \right) \right] + \sum_i \text{tr}(\sigma_i M_i)$$

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DUAL – upper bound

$$\begin{aligned} &\text{minimise} \quad \text{tr}(Y) \\ \text{subject to } & p_i \rho_i - Y + \sigma_i = 0 \quad \forall i \\ & \sigma_i \succeq 0 \quad \forall i \end{aligned}$$

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State discrimination problem from Lecture 1

$$p_1 = p_2 = \frac{1}{2}, \quad \rho_1 = |0\rangle\langle 0|, \quad \rho_2 = |+\rangle\langle +|$$

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$$|m_1\rangle = \cos\left(\frac{\pi}{8}\right) |0\rangle - \sin\left(\frac{\pi}{8}\right) |1\rangle, \quad |m_2\rangle = \sin\left(\frac{\pi}{8}\right) |0\rangle + \cos\left(\frac{\pi}{8}\right) |1\rangle, \quad M_i = |m_i\rangle\langle m_i|$$

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$$\begin{aligned} p_{\text{succ}}(\{M_1, M_2\}) &= \frac{1}{2} \text{tr}(\rho_1 M_1) + \frac{1}{2} \text{tr}(\rho_2 M_2) \\ &= \cos^2\left(\frac{\pi}{8}\right) \\ &= \frac{2 + \sqrt{2}}{4} \end{aligned}$$

This proves achievability; optimality still needs an upper bound

A dual operator reaches the same value

DUAL – upper bound

$$\begin{array}{ll} \min_{\mathbf{Y}} & \text{tr}(\mathbf{Y}) \\ \text{subject to} & \frac{1}{2} |0\rangle\langle 0| \preceq \mathbf{Y} \\ & \frac{1}{2} |+\rangle\langle +| \preceq \mathbf{Y} \end{array}$$

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$$Y^{\star} - \frac{1}{2}\rho_1 = \frac{1}{2\sqrt{2}}M_2 \succeq 0,$$

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$$\begin{array}{ll} \min_{Y^*} & \text{tr}(Y) \\ \text{subject to} & \frac{1}{2} |0\rangle\langle 0| \preceq Y^* \\ & \frac{1}{2} |+\rangle\langle +| \preceq Y^* \end{array}$$

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$$Y^* - \frac{1}{2}\rho_1 = \frac{1}{2\sqrt{2}}M_2 \succeq 0, \quad Y^* - \frac{1}{2}\rho_2 = \frac{1}{2\sqrt{2}}M_1 \succeq 0$$

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$$Y^* - \frac{1}{2}\rho_1 = \frac{1}{2\sqrt{2}}M_2 \succeq 0, \quad Y^* - \frac{1}{2}\rho_2 = \frac{1}{2\sqrt{2}}M_1 \succeq 0$$

Y^* is dual feasible, hence $\text{tr}(Y^*)$ is an upper bound.

Matching bounds prove optimality

$$\frac{2 + \sqrt{2}}{4} = p_{\text{succ}}(\{M_1, M_2\}) \leq p_{\text{succ}}^* \leq \text{tr}(\mathbf{Y}^*) = \frac{2 + \sqrt{2}}{4}$$

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But when does strong duality hold?

Slater's theorem

Slater conditions

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Two-sided Slater conditions

If the primal and dual are strictly feasible, then $\boxed{\alpha = \beta}$.

State discrimination satisfies Slater conditions

PRIMAL – achievable

DUAL – upper bound

$$M_i = \frac{\mathbb{1}}{N} \succcurlyeq 0, \quad \sum_{i=1}^N M_i = \mathbb{1}$$

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DUAL – upper bound

$$Y = \lambda \mathbb{1}, \quad \lambda > \max_i p_i$$

$$Y - p_i \rho_i \succ 0 \quad \forall i$$

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$$Y - p_i \rho_i \succ 0 \quad \forall i$$

Strong duality holds, and both optima are attained

$$p_{\text{succ}}^* := \max_{\substack{M_i \succeq 0 \ \forall i \\ \sum_i M_i = \mathbb{1}}} \sum_i p_i \text{tr}(\rho_i M_i) = \min_{Y \succeq p_i \rho_i \ \forall i} \text{tr}(Y)$$

7. Analytical applications

A useful certificate from the identity

A universal dimension bound

Theorem

For any ensemble $\{p_i, \rho_i\}_{i=1}^N$ on a d -dimensional Hilbert space,

$$p_{\text{succ}}^* \leq d \max_i p_i.$$

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Hence,

$$p_i \rho_i \preceq p_i \mathbb{1} \preceq \left(\max_i p_i \right) \mathbb{1} = Y$$

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Hence,

$$p_i \rho_i \preceq p_i \mathbb{1} \preceq \left(\max_i p_i\right) \mathbb{1} = \textcolor{violet}{Y}$$

Thus $\textcolor{violet}{Y} = \left(\max_i p_i\right) \mathbb{1}$ is dual feasible and $\text{tr}(\textcolor{violet}{Y}) = d \max_i p_i$ is an upper bound. □

Uniform priors reduce the bound to d/N

Suppose

$$p_i = \frac{1}{N} \quad i = 1, \dots, N$$

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$$\mathbf{Y} = \frac{\mathbb{1}}{N} \succeq \frac{\rho_i}{N} \quad \text{for every } i$$

$$\text{For uniform priors, } p_{\text{succ}}^* \leq \text{tr}(\mathbf{Y}) = \frac{d}{N}.$$

The most discriminable states

Theorem

Let $\{|\psi_i\rangle\}_{i=1}^N$ be pure states in $\mathbb{C}^d$. If

$$\frac{1}{N} \sum_{i=1}^N |\psi_i\rangle\langle\psi_i| = \frac{\mathbb{1}}{d},$$

then, for uniform priors, $p_{\text{succ}}^* = \frac{d}{N}$.

The most discriminable states

Proof. Set $M_i := \frac{d}{N} |\psi_i\rangle\langle\psi_i|$. Then $M_i \succeq 0$ and $\sum_i M_i = \frac{d}{N} \sum_i |\psi_i\rangle\langle\psi_i| = \mathbb{1}$, so $\{M_i\}$ is a POVM.

$$\begin{aligned} p_{\text{succ}}(\{M_i\}) &= \frac{d}{N^2} \sum_{i=1}^N \text{tr}(|\psi_i\rangle\langle\psi_i| |\psi_i\rangle\langle\psi_i|) \\ &= \frac{d}{N}. \end{aligned}$$

The most discriminable states

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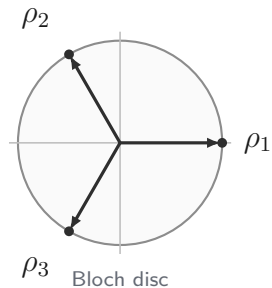
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The dual feasible point $Y = \frac{\mathbb{1}}{N}$ gives $\text{tr}(Y) = \frac{d}{N}$. □

The trine states

Let $\omega = e^{2\pi i/3}$ and, for $i = 1, 2, 3$,

$$|\psi_i\rangle = \frac{|0\rangle + \omega^{i-1}|1\rangle}{\sqrt{2}}, \quad \rho_i = |\psi_i\rangle\langle\psi_i|$$

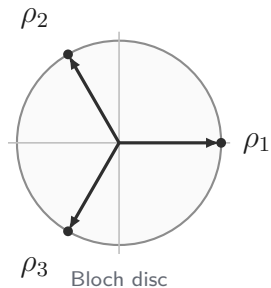


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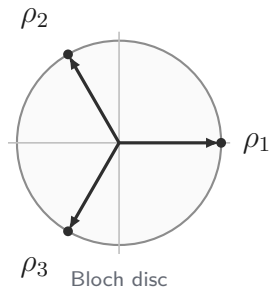
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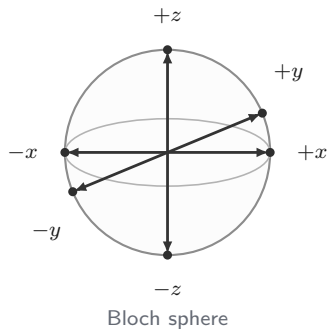
$$M_i = \frac{2}{3}\rho_i, \quad Y = \frac{\mathbb{1}}{3}, \quad p_{\text{succ}}^* = \frac{2}{3}$$



Pauli eigenstates form an optimal octahedron

For $a \in \{x, y, z\}$ and $s \in \{+1, -1\}$, let

$$\rho_{a,s} = \frac{1}{2}(\mathbb{1} + s\sigma_a)$$

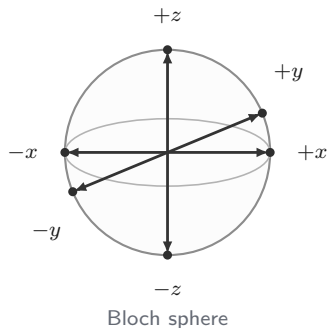


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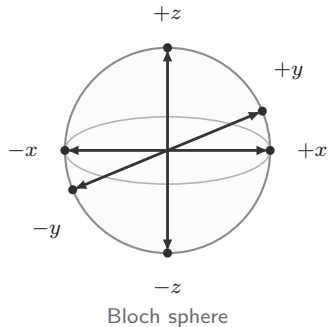
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$$M_{a,s} = \frac{1}{3}\rho_{a,s}, \quad Y = \frac{\mathbb{1}}{6}, \quad p_{\text{succ}}^* = \frac{1}{3}$$



8. Numerical solvers

From an exact model to approximate candidates

Numerical solvers

- ▶ There are *efficient* computational algorithms to solve SDPs.

Numerical solvers

- ▶ There are *efficient* computational algorithms to solve SDPs.
- ▶ For a well-posed SDP of manageable size, a solver can usually return approximate primal and dual candidates—or diagnose a problem.

Efficient means controlled scaling

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matrix data + an accuracy request ε

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and returns approximate primal and dual candidates, objective values, and diagnostics.

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Under standard regularity and finite-precision assumptions, theoretical algorithms have polynomial scaling in the encoded input size and $\log(1/\varepsilon)$

The time complexity for solving an SDP

Given: $A = A^\dagger \in \mathcal{L}(\mathbb{C}^d)$, $B = B^\dagger \in \mathcal{L}(\mathbb{C}^{d'})$

$\Lambda : \mathcal{L}(\mathbb{C}^d) \longrightarrow \mathcal{L}(\mathbb{C}^{d'})$ Hermiticity-preserving

$$\begin{array}{ll} \text{maximise} & \text{tr}(A\rho) \\ \text{subject to} & \Lambda(\rho) = B, \quad \rho \succeq 0 \end{array}$$

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Given: $A = A^\dagger \in \mathcal{L}(\mathbb{C}^d)$, $B = B^\dagger \in \mathcal{L}(\mathbb{C}^{d'})$

$\Lambda : \mathcal{L}(\mathbb{C}^d) \longrightarrow \mathcal{L}(\mathbb{C}^{d'})$ Hermiticity-preserving

$$\begin{array}{ll} \text{maximise} & \text{tr}(A\rho) \\ \text{subject to} & \Lambda(\rho) = B, \quad \rho \succeq 0 \end{array}$$

Roughly speaking,

the time needed to solve this problem to precision ε grows polynomially in d , d' , and $\log(1/\varepsilon)$.

Examples?

The primal SDP is one short function

```
using LinearAlgebra
using JuMP
import Clarabel

function state_discrimination(rho, p)
    N, d = length(rho), size(rho[1], 1)
    model = Model(Clarabel.Optimizer)
    M = [@variable(model, [1:d, 1:d] in HermitianPSDCone()) for i in 1:N]
    @constraint(model, sum(M) == I)
    @objective(
        model,
        Max,
        sum(p[i] * real(tr(rho[i] * M[i]))) for i in 1:N),
    )
    optimize!(model)
    return objective_value(model), value.(M)
end
```

The dual is just as direct

```
function state_discrimination_dual(rho, p)
    N, d = length(rho), size(rho[1], 1)
    model = Model(Clarabel.Optimizer)
    @variable(model, Y[1:d, 1:d] in HermitianPSDCone())
    @constraint(
        model,
        [i = 1:N],
        Hermitian(Y - p[i] * rho[i]) in HermitianPSDCone(),
    )
    @objective(model, Min, real(tr(Y)))
    optimize!(model)
    return objective_value(model), value.(Y)
end
```


Discriminating $|0\rangle$ from $|+\rangle$ in a few lines

```
include("state_discrimination.jl")

ket0 = [1; 0]
ket_plus = [1; 1] / sqrt(2)

rho_1 = ket0 * ket0'
rho_2 = ket_plus * ket_plus'
p_1 = 1 / 2
p_2 = 1 / 2

rho = [rho_1, rho_2]
p = [p_1, p_2]

p_primal, M = state_discrimination(rho, p)
p_dual, Y = state_discrimination_dual(rho, p)

exact = (2 + sqrt(2)) / 4

println("rho      = ", rho)
println("p        = ", p)
println("primal     = ", p_primal)
println("dual       = ", p_dual)
println("expected  = ", exact)
```

The numerical outcome

NUMERICAL

primal	0.8535533694
dual	0.8535533898
expected	0.8535533906

EXACT

$$p_{\text{succ}}^* = \frac{2 + \sqrt{2}}{4}$$

$$Y^* = \frac{1}{4} \left(\rho_1 + \rho_2 + \frac{\mathbb{1}}{\sqrt{2}} \right)$$

$$M_1 = |m_1\rangle\langle m_1|, \quad M_2 = |m_2\rangle\langle m_2|$$

Discriminating the six Pauli eigenstates

```
include("state_discrimination.jl")
sigma_x = [0 1; 1 0]
sigma_y = [0 -im; im 0]
sigma_z = [1 0; 0 -1]

rho = [Matrix((I + s * sigma) / 2)
       for sigma in (sigma_x, sigma_y, sigma_z) for s in (1, -1)]
p = fill(1 / 6, 6)

p_primal, M = state_discrimination(rho, p)
p_dual, Y = state_discrimination_dual(rho, p)

exact = 1 / 3

println("rho      = ", rho)
println("p        = ", p)
println("primal    = ", p_primal)
println("dual       = ", p_dual)
println("expected = ", exact)
```

The numerical outcome

NUMERICAL

primal	0.3333333325
dual	0.3333333330
expected	0.3333333333

The optimal POVM is not unique.

EXACT

$$M_{a,s} = \frac{1}{3}\rho_{a,s}, \quad Y = \frac{1}{6}\mathbb{1}$$

$$p_{\text{succ}}^{\star} = \frac{1}{3}$$

One exact optimal pair, independent of the solver output

Solving SDPs with computers

Language

Julia

Open source; fast.

MATLAB

Proprietary; fast.

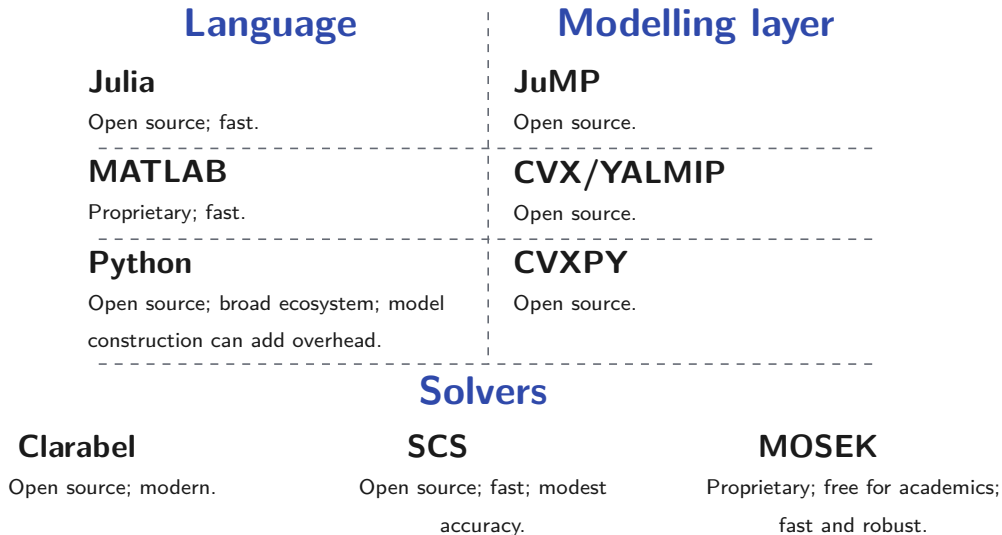
Python

Open source; broad ecosystem; model construction can add overhead.

Solving SDPs with computers

Language	Modelling layer
Julia Open source; fast.	JuMP Open source.
MATLAB Proprietary; fast.	CVX/YALMIP Open source.
Python Open source; broad ecosystem; model construction can add overhead.	CVXPY Open source.

Solving SDPs with computers



Quantum information libraries

Quantum-information libraries can simplify common tasks:

- ▶ Partial traces, partial transposition, and other convenient functions

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Julia
Ket.jl

MATLAB
QETLAB

Python
QuTiP

This is not an exhaustive list—only a few libraries I have used and liked.

Floating-point arithmetic?

Floating-point arithmetic is powerful and reliable.

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Floating-point arithmetic is powerful and reliable.

But this is not a mathematical proof!

9. Computer-assisted proofs

From numerical structure to rigorous bounds

Do you trust your computer?

Sometimes yes, sometimes no.

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Sometimes yes, sometimes no.

Floating-point numbers are convenient.

But do you trust them?

What is a floating-point number?

As in scientific notation, a normal binary floating-point number has the form

$$\underbrace{(-1)^s}_{\text{sign}} \times \underbrace{(1.b_1b_2 \dots b_k)_2}_{\text{significand}} \times \underbrace{2^e}_{\text{exponent}}$$

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$$\underbrace{(-1)^s}_{\text{sign}} \times \underbrace{(1.b_1b_2 \dots b_k)_2}_{\text{significand}} \times \underbrace{2^e}_{\text{exponent}}$$

Only finitely many values can be stored exactly.

Most inputs are rounded to a nearby representable number.

Distinct numbers can share one Float64

EXACT

$$x = 9\,007\,199\,254\,740\,992, \quad y = 9\,007\,199\,254\,740\,993$$

$$x \neq y$$

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NUMERICAL

$$\text{Float64}(x) = \text{Float64}(y) = 9\,007\,199\,254\,740\,992$$

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NUMERICAL

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A finite floating-point format cannot represent every real number distinctly

Live demonstration time!

Floating-point addition is not associative

In exact real arithmetic, addition is associative:

$$(a + b) + c = a + (b + c)$$

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But for Float64:

$$(a + b) + c = 1.0$$

but

$$a + (b + c) = 0.0$$

The solver returns primal and dual candidates

$$p_1 = p_2 = \frac{1}{2}, \quad \rho_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \rho_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Primal solver output

$$p_{\text{primal}}^{\text{num}} = 0.8535533694$$

Dual solver output

$$p_{\text{dual}}^{\text{num}} = 0.8535533898$$

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$$p_{\text{primal}}^{\text{num}} = 0.8535533694$$

$$p_{\text{dual}}^{\text{num}} = 0.8535533898$$

$$M_1^{\text{num}} \approx \begin{pmatrix} 0.853551 & -0.353555 \\ -0.353555 & 0.146449 \end{pmatrix}$$

$$Y^{\text{num}} \approx \begin{pmatrix} 0.551772 & 0.124996 \\ 0.124996 & 0.301781 \end{pmatrix}$$

$$M_2^{\text{num}} \approx \begin{pmatrix} 0.146449 & 0.353555 \\ 0.353555 & 0.853551 \end{pmatrix}$$

The variables are part of the numerical output too.

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The variables are part of the numerical output too.

Can these decimal candidates lead to an exact proof?

A recipe for a dual certificate

For the dual:

1. Round or reconstruct an exact Hermitian candidate:

$$Y^{\text{num}} \rightsquigarrow \widetilde{Y} = \widetilde{Y}^\dagger.$$

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2. Obtain a rigorous bound on every dual slack and choose an exact $\varepsilon \geq 0$ such that

$$\lambda_{\min}(\widetilde{Y} - p_i \rho_i) \geq -\varepsilon \quad \forall i.$$

Then

$$Y^{\text{cert}} := \widetilde{Y} + \varepsilon \mathbb{1} \succeq p_i \rho_i \quad \forall i.$$

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$$Y^{\text{cert}} := \widetilde{Y} + \varepsilon \mathbb{1} \succeq p_i \rho_i \quad \forall i.$$

3. Evaluate $\text{tr}(Y^{\text{cert}})$ exactly: this is a rigorous upper bound.

An exact analytical proof from numerics

- Reconstruct an exact matrix from the decimal pattern:¹

$$Y^{\text{exact}} := \frac{1}{8} \begin{pmatrix} 3 + \sqrt{2} & 1 \\ 1 & 1 + \sqrt{2} \end{pmatrix}.$$

¹Of course, this is not always possible. But sometimes, yes! It is worth doing a few checks.

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- ▶ Verify both inequalities by exact rank-one factorisations:

$$Y^{\text{exact}} - \frac{1}{2}\rho_1 = \frac{1}{2\sqrt{2}} |m_2\rangle\langle m_2| \succeq 0, \quad Y^{\text{exact}} - \frac{1}{2}\rho_2 = \frac{1}{2\sqrt{2}} |m_1\rangle\langle m_1| \succeq 0.$$

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- ▶ Evaluate the dual objective exactly:

$$p_{\text{succ}}^* \leq p_{\text{upper}}^* := \text{tr}(\mathbf{Y}^{\text{exact}}) = \frac{(3 + \sqrt{2}) + (1 + \sqrt{2})}{8} = \frac{2 + \sqrt{2}}{4}.$$

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Choose an exact $\eta \in [0, 1]$ satisfying $\eta \geq N\delta/(1 + N\delta)$, and set

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4. Evaluate the primal objective exactly to obtain a rigorous lower bound:

$$\sum_i p_i \operatorname{tr}(\rho_i M_i^{\text{cert}}) \leq p_{\text{succ}}^*.$$

Certify the primal candidate

- Reconstruct exact matrices from the decimal pattern:²

$$M_1^{\text{exact}} := \frac{1}{4} \begin{pmatrix} 2 + \sqrt{2} & -\sqrt{2} \\ -\sqrt{2} & 2 - \sqrt{2} \end{pmatrix}, \quad M_2^{\text{exact}} := \frac{1}{4} \begin{pmatrix} 2 - \sqrt{2} & \sqrt{2} \\ \sqrt{2} & 2 + \sqrt{2} \end{pmatrix}.$$

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- Verify the constraints exactly:

$$M_i^{\text{exact}} = |m_i\rangle\langle m_i| \succeq 0 \quad (i = 1, 2),$$
$$M_1^{\text{exact}} + M_2^{\text{exact}} = \frac{1}{4} \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = \mathbb{1}.$$

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- Evaluate the primal objective:

$$\begin{aligned} p_{\text{lower}}^* &:= \frac{1}{2} \text{tr}(\rho_1 M_1^{\text{exact}}) + \frac{1}{2} \text{tr}(\rho_2 M_2^{\text{exact}}) \\ &= \frac{2 + \sqrt{2}}{4} = p_{\text{upper}}^*, \end{aligned}$$

$$p_{\text{lower}}^* \leq p_{\text{succ}}^* \leq p_{\text{upper}}^* \implies \boxed{p_{\text{succ}}^* = \frac{2 + \sqrt{2}}{4}}.$$

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10. Outlook

What the certificate viewpoint unlocks

Lecture 2 in brief

- ▶ The dual problem can provide good (and tight) upper bounds.

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- ▶ The dual problem can provide good (and tight) upper bounds.
- ▶ Slater conditions ensure strong duality.
- ▶ There are efficient numerical methods.
- ▶ Numerical solutions can sometimes be turned into computer-assisted proofs.

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- ▶ State discrimination is **clearly an SDP**. For some problems, this is not at all obvious.
- ▶ Sometimes we must “**massage**” the **formulation** to recognise its SDP structure.
- ▶ If all else fails, **SDP hierarchies** may still provide systematic upper and/or lower bounds.
- ▶ This should make more sense during Jessica's talk. 🌟

SDP in quantum information

- ▶ entanglement and steering
- ▶ measurement compatibility
- ▶ Bell nonlocality and correlations in networks
- ▶ quantum marginal problem
- ▶ channel discrimination
- ▶ quantum channel capacities
- ▶ quantum metrology
- ▶ higher-order computing, indefinite causality
- ▶ ...

Want more?

- ▶ Paul Skrzypczyk and Daniel Cavalcanti, *Semidefinite Programming in Quantum Information Science*;

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Thank you! 